Full-series threshold Brier scores for the time-block forecast

Per-block and per-dataset breakdowns, Experiment A, settings 4, 5 and 7

Score report

10 August 2026

Abstract

The time-block Brier score is now computed against thresholds drawn from each dataset's *full series*, replacing the validation block's own quantile. This report gives the breakdown that a pooled lead-time table hides: the same scores resolved by block and by dataset, at two co-primary thresholds, for the three settings fitted under this campaign. Those three span persistence end to end — a short-memory control (setting 4), a near-unit-root process (setting 5) and a long-memory process (setting 7) — so the score can be read against the property it is supposed to be sensitive to. At the near-unit-root setting 5 the AR(2) advantage is not an aggregate artifact — AR(2) scores below i.i.d. in **all five blocks** and, at q_{90}, in **all ten datasets**, and the pooled contrast is significant at every threshold from q_{90} to q_{98} (AR(2)−AR(1) at q_{90}: $t\,p = 0.0002$). This reverses the conclusion of the retired campaign, which held that the Brier score was too noisy to detect what CRPS could see; under the full-series rule the Brier score alone resolves setting 5 decisively, which is why this report carries no CRPS. The control gives the other end: at setting 4, whose autocorrelation decays in six days, the pooled AR(2)−i.i.d. gap over the fifteen-day horizon is -1.0×10^{-6} at q_{90} and stays below 0.08×10^{-3} in absolute value at every one of the eleven thresholds. The score therefore moves with persistence, not with the parameter count. At the long-memory setting 7 the score is a genuine null, and §7 shows the null is *not* an artifact of threshold degeneracy: it survives at q_{90}, where the share of cells with no realised exceedance falls from 60.9% to 41.6%. What does move at setting 7 is calibration — the AR(2) predictive over-states the exceedance probability by a factor rising to four in the far tail, which is a level error the pooled score never shows.

1 Scope and design

Every number here comes from the three score caches `simstudy/output/results/scores{4,5,7}_hn.RData`, which hold the Experiment A campaign: expanding-window blocks with seams at $T_o = 50, 80, 110, 140, 170$, each forecasting $H = 15$ days; three methods (i.i.d., AR(1), AR(2)); ten datasets; five blocks. All fits carry the $\lambda \sim \mathrm{HN}(0, 20)$ prior, and the caches are complete — no cell is missing, so no clean-n selection is applied anywhere in this report and $n = 10$ throughout. The three settings are fitted under one protocol, so nothing but the generating persistence separates them.

id	label	temporal structure	persistence
4	control	AR(2) $\phi = (0.80, -0.35)$	$\rho(F) = 0.592$, $h^* = 6$ d (short memory)
5	near-unit-root	AR(2) $\phi = (0.15, 0.80)$	$\rho(F) = 0.973$, $h^* = 109$ d (short memory)
7	LM $d = 0.45$	ARFIMA$(0, 0.45, 0)$	Hurst 0.95, hyperbolic ACF (long memory)

Setting 4 is the design's negative control. Its memory horizon is $h^* = 6$ days, well inside the 15-day forecast window, so by lead 7 the AR(2) and i.i.d. predictives have converged and the comparison has nothing left to detect. A campaign that reported a gain here would be reporting an artifact of the extra parameters; the value of the setting lies in the null it is expected to produce.

Two co-primary thresholds. Tables report q_{90} and q_{95} side by side rather than q_{95} alone. They are not interchangeable: the two disagree on how many datasets favour AR(2) at every setting (10/10 vs. 9/10 at setting 5; 6/10 vs. 4/10 at setting 7; 5/10 vs. 4/10 at setting 4), and q_{90} has materially better coverage, which §7 needs. The full eleven-point grid is in the appendix.

What is not here. Setting 6 (ARFIMA $d = 0.35$) and Experiments B and C are absent. They were fitted under the default $\lambda \sim N(0, 20)$ prior, which is a different arm of the campaign, and adding them would mean mixing priors inside one table rather than re-scoring an existing cache. Their fits do survive on the machine that produced them — `simstudy/results/` holds settings 1–4 and `block1_positive_control/results/` holds the 160 block-1 fits of the guarded campaign — but every one of them carries the wrong prior for this report, and the block-1 fits carry a different protocol as well (one block, a fit-level gate, and a clean-n selection). Those experiments, and the CRPS results for all of them, remain recorded in `tex/timeblock_expABC_legacy`, which is frozen for that reason. Setting 4 was in this list until 2026-08-09, when it was refitted from scratch under the present protocol; it is now a full member of the campaign.

Brier only. This report and its sibling `tex/timeblock_brier_blockq` are Brier-only by construction. CRPS is invariant to the threshold rule, so it cannot discriminate between the two documents; and, as the abstract notes, it is no longer needed to carry the setting-5 result.

2 The full-series threshold rule

For dataset d at setting s the thresholds are the quantiles of the whole record,

$$\tau_p(d, s) \;=\; \text{quantile}\big(y[\,\cdot, \cdot, d, s\,],\, p\big), \qquad p \in \{0.90, \ldots, 0.995\},$$

fixed across every block, lead and method, and identical for the methods being compared. This is the rule the spatial simstudy already used (`code/analysis/simstudy/scores.R`), so the two studies are now commensurable.

The rule it replaced took the threshold from the validation window itself, $\text{quantile}(y_{\text{val}}, p)$. That is look-ahead — the threshold is a statistic of the very window being forecast — and it has a second, sharper defect quantified in the sibling report: it pins the realised exceedance rate at exactly $1 - p$ in every block, so no level error can move the base rate. The scorer still computes it, as `brier.lead.blockq`, and the caches record which rule produced which array in `brier_threshold_basis`; the merge gate refuses to combine caches written under different rules, so an old-rule chunk cannot silently average into a new-rule campaign. `expA_breakdown.R` re-checks that tripwire on load.

3 Per-block breakdown

Setting 5. AR(2) scores below i.i.d. in all five blocks at both thresholds. The advantage is therefore not an artifact of pooling: no single window carries it. What the breakdown also

Table 1: Experiment A, full-series threshold rule: mean Brier score by block, leads 1–15 and the remaining axis pooled. Levels in native units; the AR(2)−i.i.d. gap in units of 10^{-3} (negative favours AR(2)). Co-primary thresholds q_{90} and q_{95} side by side.

		q_{90}				q_{95}			
setting	block	i.i.d.	AR(1)	AR(2)	gap	i.i.d.	AR(1)	AR(2)	gap
4	1	0.0644	0.0654	0.0647	+0.2	0.0303	0.0310	0.0308	+0.5
	2	0.0625	0.0672	0.0661	+3.6	0.0237	0.0260	0.0253	+1.6
	3	0.1111	0.1067	0.1054	−5.6	0.0682	0.0665	0.0652	−3.1
	4	0.1243	0.1269	0.1253	+1.0	0.0691	0.0700	0.0693	+0.3
	5	0.0755	0.0755	0.0763	+0.8	0.0420	0.0420	0.0424	+0.3
5	1	0.1628	0.1619	0.1353	−27.5	0.0850	0.0841	0.0717	−13.3
	2	0.1673	0.1724	0.1463	−21.0	0.1053	0.1102	0.0992	−6.1
	3	0.0451	0.0513	0.0374	−7.7	0.0154	0.0213	0.0114	−4.0
	4	0.0525	0.0511	0.0350	−17.5	0.0283	0.0279	0.0215	−6.8
	5	0.1759	0.1573	0.1206	−55.4	0.1060	0.1044	0.0917	−14.3
7	1	0.0715	0.0741	0.0667	−4.7	0.0339	0.0360	0.0320	−1.9
	2	0.0867	0.0882	0.0955	+8.8	0.0393	0.0432	0.0458	+6.5
	3	0.1046	0.1069	0.1018	−2.8	0.0621	0.0639	0.0612	−0.9
	4	0.1279	0.1324	0.1254	−2.5	0.0785	0.0830	0.0777	−0.8
	5	0.1212	0.1200	0.1218	+0.5	0.0644	0.0652	0.0680	+3.6

exposes is how unequal the windows are — at q_{95} the i.i.d. level ranges from 0.0154 (block 3) to 0.1060 (block 5), nearly sevenfold, and at q_{90} from 0.0451 to 0.1759. That spread is the rule working as intended. A fixed threshold lets a quiet window be quiet and a turbulent one be turbulent; the block-quantile rule flattens exactly this, which is what the sibling report is about. The largest gap sits in the most turbulent window (block 5, -55.4×10^{-3} at q_{90}), i.e. the AR(2) memory pays where there is something to predict.

Setting 7. Mixed in sign — AR(2) ahead in three of five blocks at both thresholds, behind in block 2 by more than it leads anywhere. No block-level structure supports an advantage.

Setting 4. AR(2) is ahead in one block of five at both thresholds (block 3, -5.6×10^{-3} at q_{90}) and marginally behind in the other four, none of them by more than 3.6×10^{-3}. The pooled gap is -1.0×10^{-6} because one block's lead cancels four small deficits, which is what scatter around zero looks like when it is resolved by block. The contrast with setting 5 is the point of putting the two in the same table: identical protocol, identical estimator, and five aligned signs there against a cancelling five here.

4 Per-dataset breakdown

The dataset is the pairing unit of the tests in §5, so this table shows what those p-values are made of. At setting 5 and q_{90}, AR(2) wins in *all ten* datasets, with gaps from −0.0787 down to −0.0001; at q_{95} it wins in nine, the exception being dataset 10 at +0.0008. Dataset 10 is in fact the least favourable dataset in all four (setting, threshold) combinations, which is worth recording as a single consistent outlier rather than noise. Neither result depends on one or two datasets carrying the mean.

At setting 7 the count is 6/10 at q_{90} and 4/10 at q_{95} — indistinguishable from a coin, and the individual gaps are an order of magnitude smaller than setting 5's.

Table 2: Experiment A, full-series threshold rule: mean Brier score by data set, leads 1–15 and the remaining axis pooled. Levels in native units; the AR(2)−i.i.d. gap in units of 10^{-3} (negative favours AR(2)). Co-primary thresholds q_{90} and q_{95} side by side.

		q_{90}				q_{95}			
setting	data set	i.i.d.	AR(1)	AR(2)	gap	i.i.d.	AR(1)	AR(2)	gap
4	1	0.0591	0.0593	0.0638	+4.7	0.0281	0.0284	0.0301	+2.0
	2	0.0903	0.0906	0.0896	−0.6	0.0583	0.0586	0.0579	−0.4
	3	0.0923	0.0924	0.0929	+0.5	0.0441	0.0444	0.0444	+0.3
	4	0.1395	0.1399	0.1378	−1.8	0.0833	0.0837	0.0828	−0.5
	5	0.0932	0.0996	0.0929	−0.4	0.0525	0.0559	0.0525	−0.1
	6	0.0948	0.0965	0.0992	+4.4	0.0518	0.0517	0.0538	+2.0
	7	0.0378	0.0408	0.0389	+1.1	0.0128	0.0138	0.0132	+0.3
	8	0.0771	0.0737	0.0694	−7.8	0.0431	0.0407	0.0380	−5.1
	9	0.0875	0.0858	0.0859	−1.6	0.0376	0.0380	0.0378	+0.1
	10	0.1038	0.1047	0.1052	+1.4	0.0550	0.0559	0.0555	+0.5
5	1	0.0589	0.0709	0.0474	−11.5	0.0180	0.0278	0.0137	−4.3
	2	0.1837	0.1439	0.1050	−78.7	0.1151	0.1069	0.0991	−16.0
	3	0.0849	0.0800	0.0533	−31.6	0.0203	0.0192	0.0109	−9.4
	4	0.1460	0.1442	0.1238	−22.2	0.0901	0.0898	0.0850	−5.1
	5	0.1009	0.0985	0.0856	−15.3	0.0592	0.0578	0.0538	−5.3
	6	0.1078	0.1014	0.0993	−8.5	0.0673	0.0635	0.0618	−5.5
	7	0.0674	0.0664	0.0586	−8.8	0.0326	0.0321	0.0289	−3.7
	8	0.1320	0.1293	0.0966	−35.4	0.0831	0.0859	0.0628	−20.2
	9	0.1552	0.1550	0.1093	−45.9	0.0900	0.0899	0.0696	−20.4
	10	0.1704	0.1987	0.1704	−0.1	0.1045	0.1228	0.1053	+0.8
7	1	0.1566	0.1518	0.1505	−6.0	0.0900	0.0879	0.0876	−2.4
	2	0.0745	0.0764	0.0746	+0.2	0.0375	0.0384	0.0377	+0.1
	3	0.0630	0.0835	0.0626	−0.4	0.0291	0.0459	0.0300	+0.9
	4	0.0909	0.0915	0.0912	+0.3	0.0516	0.0520	0.0520	+0.4
	5	0.1252	0.1331	0.1187	−6.5	0.0704	0.0779	0.0706	+0.2
	6	0.1364	0.1259	0.1299	−6.5	0.0752	0.0743	0.0734	−1.8
	7	0.1062	0.1068	0.1123	+6.1	0.0541	0.0556	0.0625	+8.3
	8	0.1142	0.1136	0.1112	−3.0	0.0645	0.0641	0.0632	−1.3
	9	0.0855	0.0852	0.0843	−1.2	0.0435	0.0436	0.0432	−0.3
	10	0.0712	0.0755	0.0870	+15.8	0.0405	0.0428	0.0493	+8.8

Setting 4 reads the same way, 5/10 and 4/10, with the per-dataset gaps spread symmetrically about zero (at q_{90}, from -0.0078 on dataset 8 to $+0.0047$ on dataset 1). No dataset is a consistent outlier across the two thresholds in the way setting 5's dataset 10 is.

5 Paired tests

Table 3: Experiment A, full-series threshold rule: paired one-sided tests (H_1: the first method of the contrast scores lower), data-set unit $n = 10$ — the lead window is averaged first, then the five blocks. Gaps in units of 10^{-3}; $p < 0.05$ in bold.

			q_{90}			q_{95}		
setting	contrast	leads	gap	t p	W p	gap	t p	W p
4	AR(1)−i.i.d.	1–5	−0.5	0.427	0.577	−0.3	0.434	0.812
	AR(1)−i.i.d.	1–15	+0.8	0.817	0.884	+0.4	0.829	0.968
	AR(2)−i.i.d.	1–5	−0.9	0.388	0.615	−0.9	0.330	0.754
	AR(2)−i.i.d.	1–15	−0.0	0.500	0.539	−0.1	0.452	0.688
	AR(2)−AR(1)	1–5	−0.4	0.411	0.423	−0.6	0.296	0.246
	AR(2)−AR(1)	1–15	−0.8	0.231	0.312	−0.5	0.182	0.138
5	AR(1)−i.i.d.	1–5	−12.9	0.053	**0.007**	−3.7	**0.032**	**0.019**
	AR(1)−i.i.d.	1–15	−1.9	0.365	0.161	+1.6	0.737	0.423
	AR(2)−i.i.d.	1–5	−32.0	**0.004**	<0.001	−10.3	**0.001**	<0.001
	AR(2)−i.i.d.	1–15	−25.8	**0.003**	<0.001	−8.9	**0.002**	**0.002**
	AR(2)−AR(1)	1–5	−19.1	**0.003**	<0.001	−6.6	**0.008**	**0.002**
	AR(2)−AR(1)	1–15	−23.9	<0.001	<0.001	−10.5	**0.001**	<0.001
7	AR(1)−i.i.d.	1–5	−1.6	0.233	0.216	+0.4	0.634	0.312
	AR(1)−i.i.d.	1–15	+1.9	0.764	0.754	+2.6	0.912	0.935
	AR(2)−i.i.d.	1–5	−0.6	0.401	0.278	+0.5	0.650	0.688
	AR(2)−i.i.d.	1–15	−0.1	0.475	0.246	+1.3	0.836	0.652
	AR(2)−AR(1)	1–5	+0.9	0.666	0.461	+0.1	0.549	0.577
	AR(2)−AR(1)	1–15	−2.1	0.250	0.278	−1.3	0.269	0.138

Tests are one-sided at the dataset unit, $n = 10$: within a dataset the lead window is averaged first (the window is the endpoint, not a replicate axis), then the five blocks, leaving one number per dataset. This is the unit `expA_threeway.R` calls primary, and `expA_breakdown.R` reproduces that script's q_{95} output to 5×10^{-16} as a check on the aggregation.

At setting 5 every AR(2) contrast is significant at both thresholds and both lead windows, on both the t and Wilcoxon tests; the sharpest is AR(2)−AR(1) over leads 1–15 at q_{90}, $t\,p = 0.0002$. That the *second-lag* contrast is the sharpest is the expected signature of $\phi = (0.15, 0.80)$, where a single lag cannot represent the process. AR(1)−i.i.d. is correspondingly the only contrast that does not hold up over the full window: it separates over leads 1–5 (Wilcoxon $p = 0.007$ at q_{90}, 0.019 at q_{95}) but not over 1–15, where it decays to $p = 0.16$ and 0.42. One lag buys something at short lead and nothing thereafter; two lags buy throughout.

At setting 7 nothing reaches significance at either threshold, in either direction.

Setting 4 likewise separates nowhere, and it is worth recording how far from separation it is: over leads 1–15 the AR(2)−i.i.d. gap is -1.0×10^{-6} at q_{90} ($t\,p = 0.50$) and -7.7×10^{-5} at q_{95} ($t\,p = 0.45$), and the closest any contrast comes to significance is AR(2)−AR(1) over 1–15 at q_{95}, at $t\,p = 0.18$. The short-lead window is the only place the control shows even a direction —

AR(2)−i.i.d. is -0.94×10^{-3} over leads 1–5 at q_{90} against -0.001×10^{-3} over 1–15 — which is the memory horizon showing through: what little the second lag buys is spent by lead 6, exactly where h^* says it should be.

6 Coverage and degeneracy

The cost of a fixed threshold is that a quiet window may contain no exceedance at all. Where that happens the Brier score has no event to discriminate and collapses onto the predicted probability alone — it still scores calibration, but it can no longer reward resolution. The ledger below is computed from the data and the thresholds only, so it is complete regardless of which fits exist.

Table 4: Coverage and degeneracy ledger, full-series rule. "zero" is the share of (lead, data set, block) cells in which the validation window realises *no* exceedance of the fixed threshold, so the Brier score there degenerates onto climatology. "realised" is the mean realised exceedance rate and "pred" the mean predicted exceedance probability (`pexceed.mean`); predicted far below realised is a level error.

	setting 5				setting 7			
p	zero	realised	pred i.i.d.	pred AR(2)	zero	realised	pred i.i.d.	pred AR(2)
0.900	59.3%	0.1257	0.1189	0.1027	41.6%	0.1111	0.0927	0.1250
0.910	62.3%	0.1154	0.1101	0.0925	43.2%	0.1001	0.0848	0.1159
0.920	65.6%	0.1049	0.1010	0.0826	47.7%	0.0891	0.0772	0.1068
0.930	68.0%	0.0932	0.0910	0.0723	50.9%	0.0788	0.0695	0.0976
0.940	72.1%	0.0812	0.0803	0.0617	55.6%	0.0680	0.0614	0.0878
0.950	75.3%	0.0682	0.0687	0.0513	60.9%	0.0574	0.0535	0.0781
0.960	79.3%	0.0549	0.0572	0.0415	67.6%	0.0462	0.0454	0.0680
0.970	82.1%	0.0412	0.0463	0.0322	72.9%	0.0346	0.0370	0.0573
0.980	85.9%	0.0273	0.0354	0.0231	79.2%	0.0230	0.0278	0.0453
0.990	91.1%	0.0133	0.0231	0.0139	87.6%	0.0115	0.0180	0.0321
0.995	93.2%	0.0065	0.0175	0.0096	92.1%	0.0057	0.0120	0.0234

At q_{95}, 80.8% of (lead, dataset, block) cells at setting 4, 75.3% at setting 5 and 60.9% at setting 7 realise no exceedance; by $p = 0.995$ this is above 92% at all three. This is the reason q_{90} is carried as a co-primary threshold rather than relegated: at q_{90} the degenerate shares fall to 63.3%, 59.3% and 41.6%. The ordering is worth noting — the control has the *poorest* coverage of the three, so its null is also the least powerful of the three, and §7 treats it accordingly.

The predicted-rate columns are a level-error ledger. At setting 5 both methods under-predict exceedance relative to what is realised (0.1189 and 0.1027 against 0.1257 at q_{90}), AR(2) slightly more so. Setting 7 inverts, and this is the more interesting reading: AR(2) *over*-predicts throughout (0.1250 against a realised 0.1111 at q_{90}), and the discrepancy widens with the threshold until at $p = 0.995$ it predicts 0.0234 against a realised 0.0057 — a factor of four. Under long memory the fitted AR(2) puts too much mass in the upper tail. A pooled Brier score cannot show this; the ledger can. Setting 4 sits between the two and is mild in both directions: both methods over-predict slightly at q_{90} (0.1022 for i.i.d. and 0.1082 for AR(2) against a realised 0.0957), and the ratio never reaches 1.8 anywhere on the grid, against setting 7's four.

7 Is the setting-7 tie signal, or degeneracy?

A null on a score that has degenerated on most of its cells is not evidence of no difference. Setting 7 must therefore be checked before its tie is believed, and the check is available inside the Brier score itself: compare thresholds whose coverage differs materially.

Moving from q_{95} to q_{90} cuts setting 7's degenerate share from 60.9% to 41.6% — the best-covered threshold on the grid. The tie survives it. The pooled gap is -0.14×10^{-3} ($t\,p = 0.475$), the dataset count 6/10, the block count 3/5; every AR(2) contrast in §5 is non-significant at q_{90} exactly as at q_{95}. The appendix sweep shows the same across all eleven thresholds: the setting-7 gap is small and *changes sign* at q_{91}, drifting mildly in i.i.d.'s favour as p rises, with the dataset count decaying monotonically from 6/10 to 2/10. That drift is not evidence for i.i.d. either — it tracks the over-prediction documented in §6, which penalises AR(2) precisely where realised exceedance is rarest.

The conclusion is that setting 7's null is a property of the forecasts, not of the scoring rule. The contrast with setting 5 makes the point sharply: setting 5 is significant at every threshold from q_{90} to q_{98} and only loses significance at q_{99} and above, where 91% of cells are degenerate. Degeneracy does destroy the Brier score's power — visibly, at the top of the grid — but at q_{90} and q_{95} there is enough left to separate setting 5 comfortably, so the failure to separate setting 7 is informative.

The control tightens the same argument from the other side, and it does so on matched coverage. At q_{90} setting 4 leaves 63.3% of cells degenerate against setting 5's 59.3% — close enough that the two nulls cannot be attributed to different amounts of surviving signal — yet setting 5 separates at $t\,p = 0.0002$ while setting 4's gap is -1.0×10^{-6}. Nor is the control's flatness confined to one threshold: the gap stays under 0.08×10^{-3} in absolute value across all eleven, and the count of datasets favouring AR(2) sits at 5 or 4 throughout, which is the pattern a genuinely absent effect produces rather than a suppressed one. Read together, the three settings order themselves by persistence exactly as the design predicts: nil at $h^* = 6$ days, decisive at $h^* = 109$, and, under hyperbolic decay, a null that §6 attributes to a level error rather than to the absence of memory.

8 Provenance

Caches: `simstudy/output/results/scores4_hn.RData`, `scores5_hn.RData`, `scores7_hn.RData`, all three with `brier_threshold_basis = "full_series"` and `hn_prior = TRUE`, written by `simstudy/scores.R`. The rule changed in commit `46628f7`; the settings-5 and -7 outputs were refreshed under it in `638f85a`, and setting 4 was fitted and scored under it in `072c07b`.

An untagged `scores4.RData` also sits in the same directory. It is the retired campaign's setting-4 cache — $N(0, 20)$ prior, two methods, the block-quantile rule, twenty datasets — and it is not the file this report reads. The `_hn` suffix is the only thing separating them on disk, which is the reason for the tagging convention described next.

The `_hn` suffix is the lambda-prior arm. Since 2026-08-05 every per-arm artifact carries it — fits in `results_hn/`, caches in `scores<S>_hn.RData` — because the formal experiment runs under either $\lambda \sim \mathrm{HN}(0, 20)$ (`--hn`) or $\lambda \sim N(0, 20)$, and before tagging the two arms silently overwrote each other. This report is the HN arm; the N arm has not been fitted.

Tables and their `_body.tex` fragments: `Rscript expA_breakdown.R --prior=hn "--settings=(4,5,7)"`, which reads only the three caches and `simdata.RData` and writes every table this report `\input`s. Each fragment records its arm in a header comment. No number in this document is transcribed by hand. The script asserts the threshold basis and the absence of NA cells on load, and cross-checks its q_{95} tests against `expA_tests_{5,7}.csv`.

Sibling report on the retired rule, including the controlled contrast between the two rules on these same fits: `tex/timeblock_brier_blockq`. Frozen record of the earlier campaign and of all CRPS results: `tex/timeblock_expABC_legacy`.

A The full threshold grid

Table 5: Experiment A, full-series threshold rule: the whole threshold grid, leads 1–15. Gap = AR(2)−i.i.d. in units of 10^{-3}, data-set unit $n = 10$. "won" counts the data sets (of 10) and blocks (of 5) in which AR(2) scores below i.i.d.

setting	p	i.i.d.	AR(1)	AR(2)	gap	t p	W p	ds won	blk won
4	0.900	0.0875	0.0883	0.0875	−0.0	0.500	0.539	5/10	1/5
	0.910	0.0795	0.0802	0.0795	+0.0	0.500	0.577	5/10	1/5
	0.920	0.0712	0.0719	0.0712	−0.0	0.494	0.577	5/10	1/5
	0.930	0.0632	0.0638	0.0632	−0.0	0.478	0.652	5/10	1/5
	0.940	0.0553	0.0558	0.0552	−0.1	0.461	0.652	5/10	1/5
	0.950	0.0467	0.0471	0.0466	−0.1	0.452	0.688	4/10	1/5
	0.960	0.0383	0.0386	0.0382	−0.1	0.444	0.688	4/10	1/5
	0.970	0.0292	0.0295	0.0292	−0.1	0.442	0.722	4/10	1/5
	0.980	0.0200	0.0202	0.0200	−0.0	0.489	0.722	4/10	1/5
	0.990	0.0096	0.0098	0.0096	+0.0	0.598	0.784	4/10	1/5
	0.995	0.0048	0.0049	0.0048	+0.0	0.531	0.652	4/10	2/5
5	0.900	0.1207	0.1188	0.0949	−25.8	**0.003**	$<$0.001	10/10	5/5
	0.910	0.1119	0.1110	0.0896	−22.3	**0.002**	**0.002**	9/10	5/5
	0.920	0.1026	0.1025	0.0837	−18.9	**0.002**	**0.002**	9/10	5/5
	0.930	0.0919	0.0924	0.0764	−15.5	**0.002**	**0.002**	9/10	5/5
	0.940	0.0806	0.0817	0.0684	−12.2	**0.002**	**0.002**	9/10	5/5
	0.950	0.0680	0.0696	0.0591	−8.9	**0.002**	**0.002**	9/10	5/5
	0.960	0.0550	0.0569	0.0490	−6.0	**0.004**	**0.002**	9/10	5/5
	0.970	0.0417	0.0436	0.0380	−3.8	**0.009**	**0.002**	9/10	5/5
	0.980	0.0280	0.0296	0.0260	−2.0	**0.048**	**0.003**	9/10	5/5
	0.990	0.0140	0.0152	0.0132	−0.9	0.169	0.065	8/10	4/5
	0.995	0.0072	0.0082	0.0067	−0.6	0.188	0.065	8/10	4/5
7	0.900	0.1024	0.1043	0.1022	−0.1	0.475	0.246	6/10	3/5
	0.910	0.0933	0.0954	0.0935	+0.2	0.548	0.385	6/10	3/5
	0.920	0.0839	0.0862	0.0845	+0.6	0.636	0.423	5/10	3/5
	0.930	0.0749	0.0774	0.0758	+0.9	0.699	0.423	5/10	3/5
	0.940	0.0653	0.0679	0.0664	+1.1	0.767	0.461	5/10	3/5
	0.950	0.0556	0.0583	0.0569	+1.3	0.836	0.652	4/10	3/5
	0.960	0.0453	0.0479	0.0467	+1.4	0.886	0.722	4/10	3/5
	0.970	0.0343	0.0368	0.0359	+1.5	0.921	0.884	3/10	2/5
	0.980	0.0231	0.0253	0.0245	+1.4	0.938	0.920	3/10	1/5
	0.990	0.0117	0.0132	0.0129	+1.2	0.940	0.958	3/10	0/5
	0.995	0.0058	0.0068	0.0068	+1.0	0.938	0.993	2/10	0/5

Pooled over leads 1–15 at the dataset unit. Setting 5 is significant on both tests from q_{90} through q_{98} and fails only at q_{99} and q_{995}, where the degenerate share exceeds 91%; the gap decays smoothly with p as the base rate does. Setting 7 never separates at any threshold. Setting 4 never separates either, and its gap never leaves the interval $[-0.08, +0.01] \times 10^{-3}$, changing sign three times inside it; the smallest $t\,p$ on its eleven rows is 0.44.

B The second-lag contrast across the grid

The sweep above is AR(2) against i.i.d., which mixes what the first lag buys with what the second adds. Table 6 isolates the second lag by sweeping AR(2) against AR(1) over the same grid. At setting 5 the contrast separates from q_{90} through q_{98}, in 10/10 data sets and 5/5 blocks throughout, and fails only at q_{99} and q_{995}, where more than 91% of cells carry no realised exceedance. That the second-lag contrast is significant on its own, and over a wider band of the tail than AR(1)−i.i.d. manages anywhere, is what identifies the second lag rather than temporal pooling in general as the source of the setting-5 result. Setting 4 never separates ($t\,p$ between 0.15 and 0.23); setting 7 never separates either, though its gap is negative at all eleven levels and 7/10 data sets favour AR(2) over AR(1) at every level up to q_{97} — a faint and consistent preference that the pooled test cannot resolve at $n = 10$.

Table 6: Experiment A, full-series threshold rule: the whole threshold grid, leads 1–15. Gap = AR(2)−AR(1) in units of 10^{-3}, data-set unit $n = 10$. "won" counts the data sets (of 10) and blocks (of 5) in which AR(2) scores below AR(1).

setting	p	i.i.d.	AR(1)	AR(2)	gap	t p	W p	ds won	blk won
	0.900	0.0875	0.0883	0.0875	−0.8	0.231	0.312	5/10	4/5
	0.910	0.0795	0.0802	0.0795	−0.8	0.217	0.246	6/10	4/5
	0.920	0.0712	0.0719	0.0712	−0.7	0.209	0.216	6/10	4/5
	0.930	0.0632	0.0638	0.0632	−0.7	0.199	0.161	7/10	4/5
	0.940	0.0553	0.0558	0.0552	−0.6	0.197	0.138	7/10	4/5
4	0.950	0.0467	0.0471	0.0466	−0.5	0.182	0.138	7/10	4/5
	0.960	0.0383	0.0386	0.0382	−0.4	0.177	0.116	8/10	4/5
	0.970	0.0292	0.0295	0.0292	−0.3	0.160	0.116	8/10	4/5
	0.980	0.0200	0.0202	0.0200	−0.2	0.155	0.116	8/10	4/5
	0.990	0.0096	0.0098	0.0096	−0.1	0.150	0.080	8/10	4/5
	0.995	0.0048	0.0049	0.0048	−0.1	0.156	0.080	8/10	3/5
	0.900	0.1207	0.1188	0.0949	−23.9	<0.001	<0.001	10/10	5/5
	0.910	0.1119	0.1110	0.0896	−21.4	<0.001	<0.001	10/10	5/5
	0.920	0.1026	0.1025	0.0837	−18.8	<0.001	<0.001	10/10	5/5
	0.930	0.0919	0.0924	0.0764	−16.0	<0.001	<0.001	10/10	5/5
	0.940	0.0806	0.0817	0.0684	−13.3	<0.001	<0.001	10/10	5/5
5	0.950	0.0680	0.0696	0.0591	−10.5	**0.001**	<0.001	10/10	5/5
	0.960	0.0550	0.0569	0.0490	−7.9	**0.003**	<0.001	10/10	5/5
	0.970	0.0417	0.0436	0.0380	−5.6	**0.008**	<0.001	10/10	5/5
	0.980	0.0280	0.0296	0.0260	−3.6	**0.029**	<0.001	10/10	5/5
	0.990	0.0140	0.0152	0.0132	−2.1	0.084	**0.019**	9/10	4/5
	0.995	0.0072	0.0082	0.0067	−1.6	0.095	**0.032**	8/10	4/5
	0.900	0.1024	0.1043	0.1022	−2.1	0.250	0.278	7/10	3/5
	0.910	0.0933	0.0954	0.0935	−1.9	0.252	0.278	7/10	3/5
	0.920	0.0839	0.0862	0.0845	−1.7	0.264	0.278	7/10	3/5
	0.930	0.0749	0.0774	0.0758	−1.6	0.264	0.216	7/10	3/5
	0.940	0.0653	0.0679	0.0664	−1.5	0.265	0.161	7/10	3/5
7	0.950	0.0556	0.0583	0.0569	−1.3	0.269	0.138	7/10	3/5
	0.960	0.0453	0.0479	0.0467	−1.2	0.276	0.161	7/10	3/5
	0.970	0.0343	0.0368	0.0359	−0.9	0.292	0.188	7/10	3/5
	0.980	0.0231	0.0253	0.0245	−0.7	0.302	0.348	5/10	3/5
	0.990	0.0117	0.0132	0.0129	−0.4	0.355	0.312	5/10	3/5
	0.995	0.0058	0.0068	0.0068	−0.1	0.463	0.385	5/10	3/5